

Case 46: Wildland-to-urban WU-E transition and isolated receivers

1 Verification purpose

Case ID: CASE46_WUT. This case verifies the discrete wildland-to-urban (WTU) transition of WU-E at a contiguous interface and at an isolated urban receiver. Its primary oracle is heat/FTP ignition with a finite exposure time. The seven-run matrix also characterizes the optional fireline-intensity shortcut and is intended to expose reversed fuel eligibility, excessive reach, topology dependence, a missing heat path, or an incorrect arrival timestamp. The scientific verdict is separate from workflow state: before execution the former is NOT EVALUABLE and the latter is NOT RUN.

The implementation under test is pinned to a2dfbcdf72209733c000e5d3431e44723e15efea. The five WU-E source files listed under `reviewed_file_sha256` in `case.yaml` were byte-identical to that revision during review. Dump behavior was reviewed from the pinned `elmfire_io.f90` git blob; its exact SHA-256 is recorded in `case.yaml`. No byte-identity claim is made for that file’s separately modified working-checkout copy. Runtime scripts do not inspect a source checkout.

The optional-shortcut preflight is honestly NOT EVALUABLE. The `namelist` assigns the value 2 to `INTERFACE_MODEL_TYPE`, but the active branch tests integer `CRITICAL_HF_WUI==2`; no deterministic assignment of that integer was found. The similarly named real `CRITICAL_HF_WUI` is a different variable. The primary heat/FTP verdict does not depend on this selector. The optional shortcut rows become PASS or FAIL only when a content-bound preflight demonstrates a true selector gate, required and deterministically assigned value 2, and a nonempty reviewed assignment location.

Reading the notation. WU-E denotes the Wildland–Urban ELMFIRE model. HRR is heat-release rate; HRRPUA is heat-release rate per unit area. DFC means direct flame contact, TOA means time of arrival, and ROS means rate of spread. FTP is the material fire-thermal property used in the intended heat-to-spread law.

Current result: FAIL. 7 of 7 required variants have evaluable evidence. Workflow: **COMPLETE.**

2 Mathematical formulation and numerical implementation

The intended WTU path deposits DFC and radiation from a burning wildland cell onto an urban receiver, accumulates

$$Q_j(t + \Delta t) = Q_j(t) + [q''_{DFC,j}(t) + q''_{rad,j}(t)]\Delta t,$$

and ignites after the fixed material threshold $F_{TP,crit}$ is met. Thus a sufficiently heated target must have a finite delay $\tau_{WTU} = t_{ign,target} - t_{ign,source} > 0$. The same heat/FTP law then produces urban ROS; no 30,000-kJ branch is part of this oracle.

As an optional implementation extension, for source (i_s, j_s) and target (i_t, j_t) , the reviewed source condition in `build/source/elmfire_spread_rate.f90:665–678` is

$$r = \sqrt{(i_t - i_s)^2 + (j_t - j_s)^2}, \quad \mathcal{I} = (f_s \neq 91) \wedge (f_t = 91) \wedge (r \leq 1) \wedge (\text{FLIN}_s > 1000).$$

The comparison is strict. A true branch sets the target’s WTU flag and skips the ordinary DFC and radiation calculation for that target. In both Runge–Kutta stages, `build/source/elmfire_level_set.`

f90:2549–2595 forces its $\phi = -1$. The newly burned-cell loop then records $\text{TOA} = T$. Initialization assigns every input $\phi \leq 0$ source cell TOA zero.

The FBFM01 source has five-percent dead one-hour moisture, zero wind, and zero slope. The input-design calculation uses

$$t_r = 384/\sigma, \quad \text{FLIN} = t_r I_R R(0.3048),$$

then searches adjacent IEEE-754 binary32 adjustment values so the reviewed default-REAL operation order brackets 999, 1000, and 1001 kW m⁻¹. This is only an input oracle; acceptance uses the ELMFIRE flin raster.

The peer-reviewed WU–E publications describe interface heat transfer and heat/FTP spread [1, 2]. The supplied thesis supports this scaling on PDF page 53 (printed page 31), equations (2.42)–(2.43), and gives vegetation heat-release context on PDF page 64 (printed page 42), equations (2.81)–(2.83). The intended equations are documented directly here. Neither published source supplies the special 1000 kW m⁻¹ branch, selector linkage, forced ϕ , or dump-name oracle; those come exclusively from the pinned source.

3 Derivation of expected behavior

For the primary oracle, positive DFC-plus-radiation must be observed at the sufficient-heat adjacent and isolated receivers and ignition must occur after a positive exposure delay. The initial-wildland-source HRR suppression is diagnostic, but later propagation within the vegetative patch may create new heat-emitting sources. The complete receiver history, not the initial-source state alone, determines this observation.

For the optional shortcut, the contiguous variants request source FLIN of 999, 1000, and 1001 kW m⁻¹ at an immediately adjacent FBFM91 band. Only 1001 may transition. Isolated targets test 999 and 1001 at one-cell distance and 1001 at two-cell distance; only the one-cell 1001 target may transition. A zero-adjustment target is the spontaneous-ignition control. Thus the exact boolean oracle is

$$(0, 0, 1)_{\text{contiguous}}, \quad (0, 1, 0)_{\text{isolated}}, \quad 0_{\text{zero}}.$$

Both eligible transitions occur during the first update at $T = 0$: the heat loop sets the WTU flag, both RK stages force the target negative, and the newly-burned loop assigns TOA to the current solver time. Since initial source TOA is also zero, the exact expectation is

$$t_u = 0 \text{ s}, \quad t_s = 0 \text{ s}, \quad \tau_{WTU} = t_u - t_s = 0 \text{ s}.$$

Every ineligible target must retain terminal $\phi > 0$ and nodata TOA. The evaluator requires $\phi \leq 0$ and finite TOA to agree; either observable alone cannot create an ignition result.

4 Verification metrics and acceptance criteria

All seven variants must have a current exit-zero receipt binding the exact runtime input manifest, variants/expected.json, all seven case-local variant metadata files, variants/variant_ids.txt, the selector preflight, executable absolute path/hash/version, captured stdout/stderr, and a complete post-exit output-file snapshot. The same executable identity must appear in every receipt, and its current file hash must still match. Stale or altered artifacts are rejected.

The dump ledger is exact, not merely terminal: 61 rows have indices 1 through 61, times 0 through 60 s, and only row 61 is final. This follows the pinned loop, which dumps at current $T = 0, \dots, 59$ before

incrementing time and then makes a final dump after reaching 60 s. Final-only FLIN and TOA names use the rounded time stamp _0000060; every-dump PHI names use the separate dump index, so terminal PHI is _d0000061. All 61 canonical PHI GeoTIFFs are required. The terminal three rasters must be one-band, aligned 25-by-25 EPSG:32610 grids with the exact nodata contract and finite non-nodata values. Captured stdout must not report a propagation stall.

Metric	Observable	Acceptance rule
Evidence/ledger	receipts, hashes, logs, 61 dump rows and terminal names	all exact for 7/7 variants
Source realization	source-mask FLIN and TOA	max FLIN error ≤ 0.5 kW m ⁻¹ ; every source TOA zero
State consistency	designated-target terminal PHI and TOA	$\phi \leq 0$ iff TOA finite
Intended WTU heat	target DFC+radiation histories	positive for sufficient-heat receivers
Intended ignition delay	target TOA minus source TOA	finite and greater than zero
Contiguous shortcut matrix	below/equal/above target state	false, false, true when selector is valid
Isolated matrix	one-cell insufficient/sufficient and two-cell sufficient	false, true, false
First update	eligible target TOA and target-source delay	both exactly zero within 10 ⁻⁸ s
Zero control	zero-adjustment target PHI and TOA	$\phi > 0$, TOA nodata

The two intended heat/FTP metrics are verdict-bearing whenever complete run evidence exists. Shortcut-attribution metrics remain NOT EVALUABLE unless the selector preflight passes; they do not control the primary case decision.

5 Simulation configuration and predicted outcome

Parameter	Value	Rationale
Grid	25 × 25, 10 m, EPSG:32610	at least 10 cells per dimension; edge-buffered targets
Fuel topology	FBFM01 source, FBFM91 targets, FBFM93 background	explicit vegetative/urban eligibility and isolation
Source requests	0, 999, 1000, 1001 kW m ⁻¹	zero control and strict threshold bracket

Receiver distance	1 or 2 cells	equality boundary and reach control
Time	start 0, fixed step/max 1, stop 60 s	exact first-update and terminal ledger oracle
Output	every-step PHI; final FLIN and TOA	direct state, intensity, and time observables
Dormant ignition	one point at 61 s	keeps ALREADY_IGNITED false through the run, preventing the vacuous zero-ignition stall test without changing 0–60 s physics
Other mechanisms	spotting, crown, randomness off	removes alternate ignition mechanisms

Figure 1 is generated directly from current inputs and is not a simulation result.

Prepared fuels and ignition
 Green: wildland; red: urban; gray: nonburnable

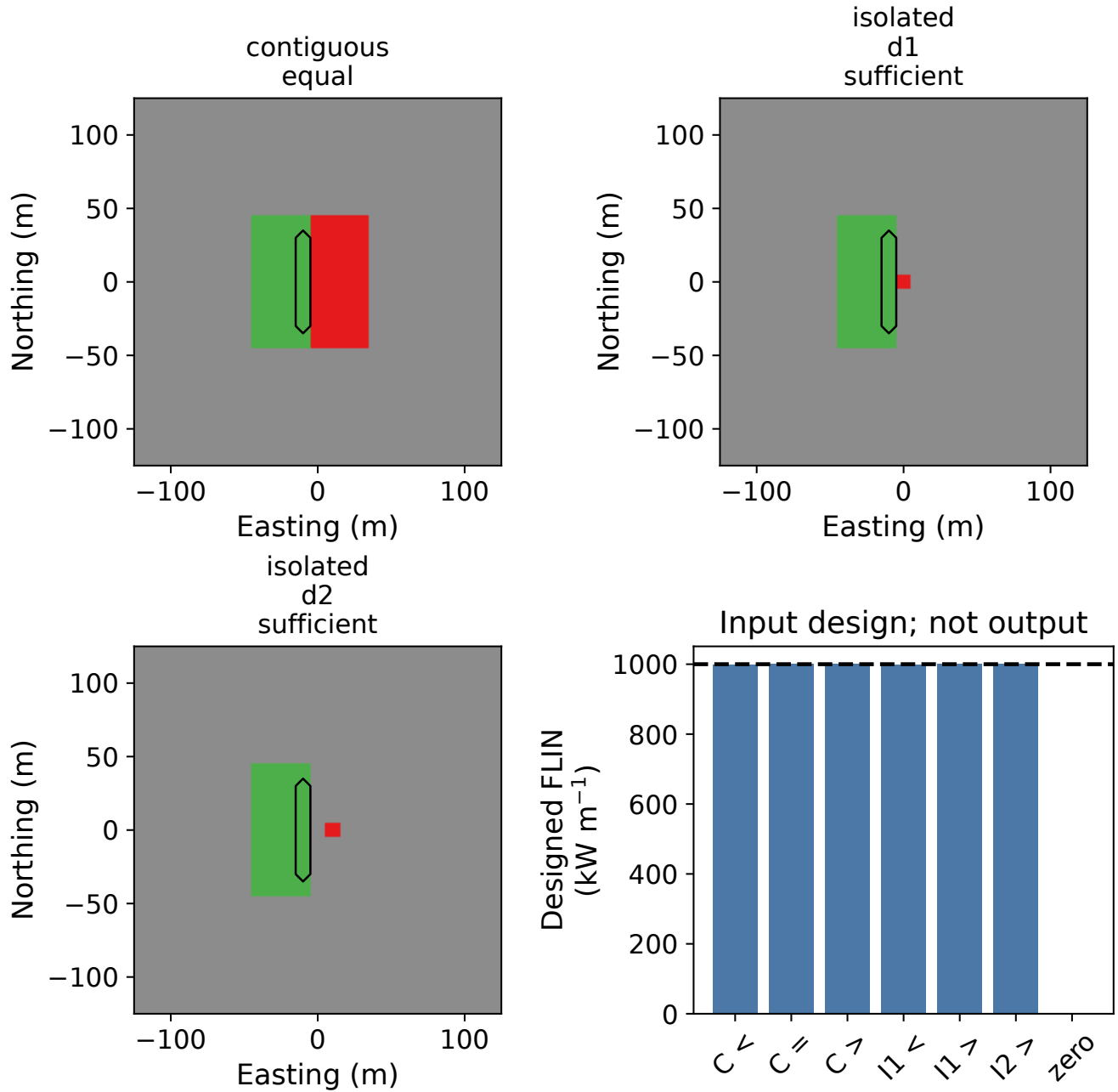


Figure 1: Prepared contiguous and isolated fuel topologies and binary32 source-FLIN design. Green is FBFM01, red is FBFM91, gray is FBFM93, and the black contour marks the seven-cell initial source.

6 Actual simulation results

Workflow status is **COMPLETE**; 7 of 7 variants have complete current evidence. Selector status is **NOT EVALUABLE**. The generated reason is:

One or more intended heat/FTP transition checks failed; optional FLIN-shortcut diagnostics do not control this decision.

The executed version is ELMFIRE 2025.1002. The full executable identity and the runtime and oracle hashes are preserved in the case-local metrics and completion receipts, so the prose need not repeat long hash strings.

Exact primary evidence is rooted at `variants/expected.json`, `variants/run_fingerprints.json`, and `outputs/source_selector_preflight.json`. Each variant has a completion receipt and captured standard-output log; terminal FLIN/TOA use `_0000060.tif`, and terminal PHI uses `_d0000061.tif`. The generated metrics JSON records the resolved path for every selected raster, receipt, log, and dump ledger.

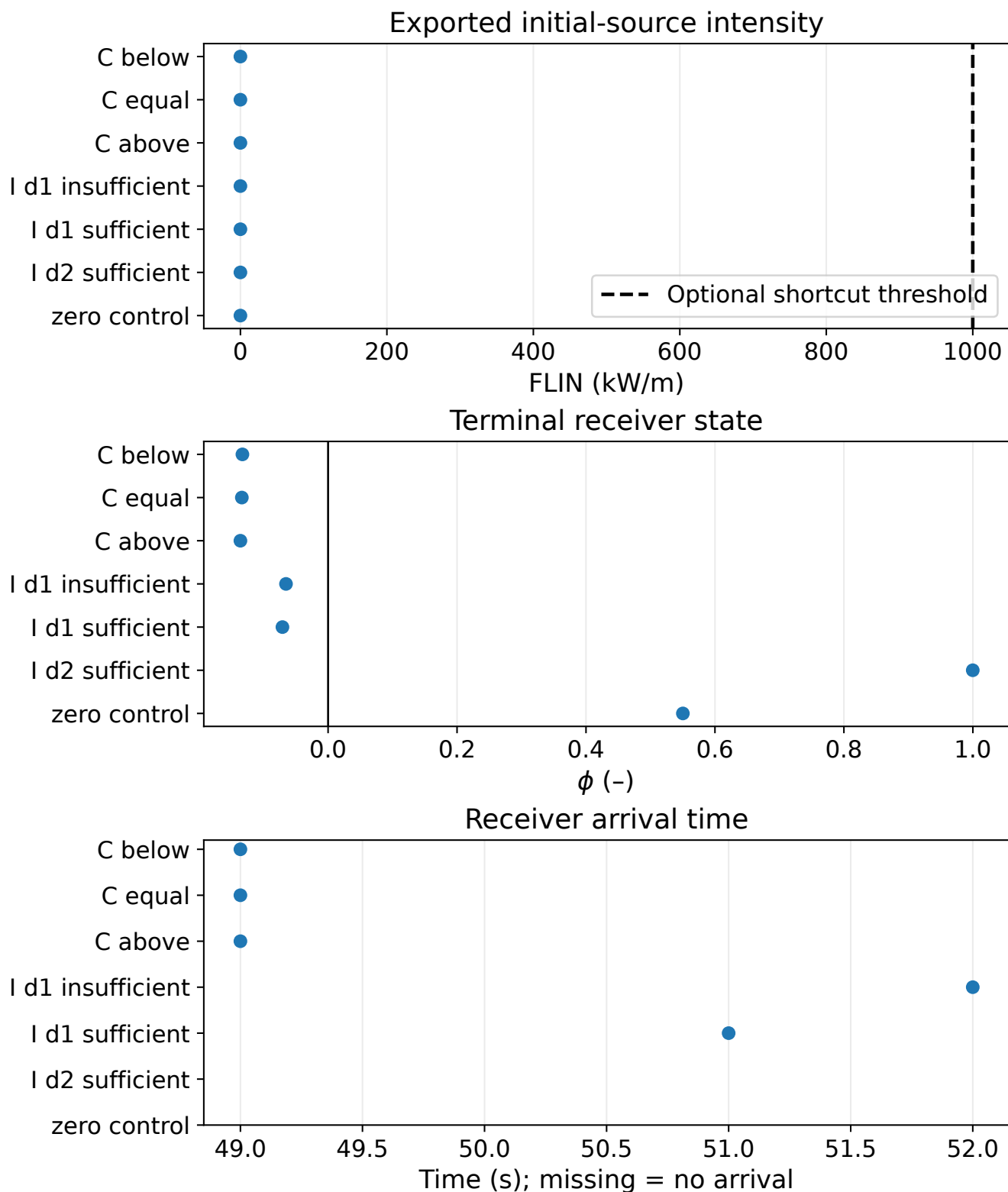


Figure 2: Current black-box source FLIN, terminal target PHI, and target TOA. C denotes contiguous targets and I isolated targets. Missing TOA means no arrival. Only shortcut attribution depends on the selector preflight; the primary checks remain evaluable.

7 Calculated metrics and verification decision

Evidence and exact dump ledger — PASS

Criterion	61 rows: indices 1..61, times 0..60 s, only row 61 final
Observed	7/7 exact; FLIN/TOA time stamp 0000060; PHI dump index d0000061

Source fireline-intensity setup — FAIL

Criterion	maximum absolute error ≤ 0.5 kW/m and source TOA=0 s
Observed	<i>Units: kW/m.</i> max abs error kw m: 1001 all source toa zero: Yes

Terminal PHI/TOA state consistency — PASS

Criterion	$\text{PHI} \leq 0$ iff TOA is finite at each designated target
Observed	contiguous below: Yes contiguous equal: Yes contiguous above: Yes isolated d1 insufficient: Yes isolated d1 sufficient: Yes isolated d2 sufficient: Yes zero control: Yes

Intended W-to-U receiver heat exposure — PASS

Criterion	positive DFC+radiation at sufficient-heat receivers
Observed	<i>Units: kW/m².</i> contiguous above: 84.5208 isolated d1 sufficient: 61.9232

Intended W-to-U finite ignition delay — PASS

Criterion	finite target TOA with delay > 0 s
Observed	<i>Units: s.</i> contiguous above: 49 isolated d1 sufficient: 51

Contiguous strict-threshold matrix — NOT EVALUABLE

Criterion	below=false, equal=false, above=true
Observed	contiguous below: Yes contiguous equal: Yes contiguous above: Yes

Isolated distance/threshold matrix — NOT EVALUABLE

Criterion	d1-insufficient=false, d1-sufficient=true, d2-sufficient=false
Observed	isolated d1 insufficient: Yes isolated d1 sufficient: Yes isolated d2 sufficient: No

First-update transition timing — NOT EVALUABLE

Criterion	eligible target TOA=0 s and target-source delay=0 s
Observed	<i>Units: s.</i> contiguous above: target toa s: 49 target source delay s: 49 isolated d1 sufficient: target toa s: 51 target source delay s: 51

Zero-source control — PASS

Criterion	target remains PHI>0 with nodata TOA
Observed	target phi: 0.550248 target toa s: Not available

The authoritative scientific status is **FAIL**. A prepared but unexecuted case is NOT EVALUABLE / NOT RUN. Complete evidence under the selector defect can still evaluate the primary heat/FTP capability; only shortcut-specific rows remain NOT EVALUABLE.

8 Technical interpretation

The current complete run fails the source-intensity setup check. The input oracle specifies head-fire FLIN, whereas the terminal raster exports surface-plus-canopy intensity from the initially burned-node list. That export is not the same observation as the head-fire intensity assigned to the WUI source list. A reliable threshold comparison requires a source diagnostic that exposes the intensity actually consumed by the transition branch.

The receiver heat and positive-delay checks can still pass. Although the initial wildland cells have suppressed transient HRR, the surrounding patch can spread and create later heat-emitting cells. Positive receiver heat does not therefore contradict initial-source suppression. Nor does a positive arrival delay alone establish the complete intended accumulated-heat/FTP law.

The optional shortcut has a separate selector defect: the exposed namelist value does not assign the integer tested by the branch. Its attribution rows remain NOT EVALUABLE. The primary heat, delay, state, source-setup, and control rows retain their own decisions; they are not all demoted because the optional selector is unavailable.

9 Scope and limitations

This case covers one wildland fuel/moisture state, one cell size, a 60-s window, contiguous and isolated targets, the strict intensity bracket, distance boundary, state/timing update, and zero control. It does not validate the physical choice of 1000 kW m^{-1} , urban-to-wildland spread, community propagation or convergence. Executable path/hash/version prove that one binary produced all accepted outputs but do not prove that binary was built from a2dfbcd72209733c000e5d3431e44723e15efea; the commit remains the documented implementation provenance. A source repair requires a newly reviewed, content-bound PASS preflight before this unchanged strict matrix can yield a shortcut-specific PASS or FAIL.

References

- [1] D. M. J. Purnomo et al., “Reconstructing modes of destruction in wildland–urban interface fires using a semi-physical level-set model,” *Proceedings of the Combustion Institute*, 40 (2024), 105755. [doi:10.1016/j.proci.2024.105755](https://doi.org/10.1016/j.proci.2024.105755).
- [2] D. M. J. Purnomo et al., “Integrating an urban fire model into an operational wildland fire model to simulate one dimensional wildland–urban interface fires: a parametric study,” *International Journal of Wildland Fire*, 33 (2024), WF24102. [doi:10.1071/WF24102](https://doi.org/10.1071/WF24102).